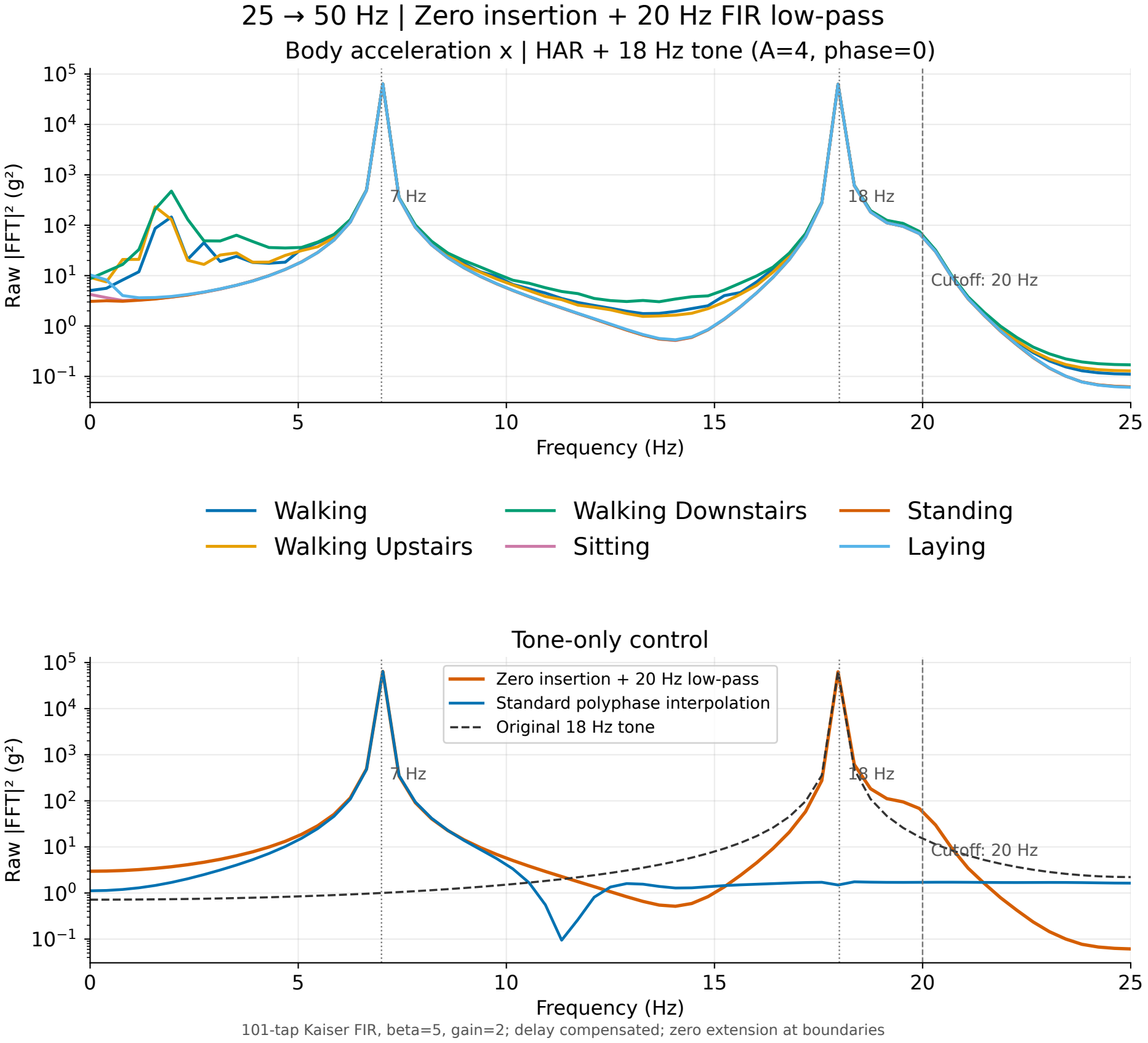
25 → 50 Hz | Zero insertion + 20 Hz FIR low-pass
Body acceleration x | HAR + 18 Hz tone (A=4, phase=0)
Raw |FFT|² (g²)
Frequency (Hz)
7 Hz
18 Hz
Cutoff: 20 Hz
Walking
Walking Upstairs
Walking Downstairs
Sitting
Standing
Laying
Tone-only control
Zero insertion + 20 Hz low-pass
Standard polyphase interpolation
Original 18 Hz tone
101-tap Kaiser FIR, beta=5, gain=2; delay compensated; zero extension at boundaries